Mingyang (Yang) Liu

Quantitative Researcher · Independent Quantitative Research & AI/ML Consultant

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Imperial College London

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Legal name: Yang Liu (刘杨); Mingyang (刘明杨) is the adult name taken at eighteen by family tradition; some listings use Yang Liu.

Selected Quantitative Research Capabilities

- **Equity signals and portfolio construction:** built characteristic-based and latent-factor models for U.S. stock-return prediction and tested them in rolling out-of-sample portfolios; traced how regularisation and signal-library choices change holdings and returns under transaction-cost scenarios.
- **Python tools:** wrote `naipca`, a Python toolkit for IPCA-family factor estimation with a companion diagnostics package, used for rolling monthly model fits; helped build the Python research and backtesting framework of the Imperial Student Investment Fund.
- **Forecast evaluation:** built a workflow that compares forecasting models on the same forecast origins, tests whether differences in forecast accuracy are statistically significant (Diebold–Mariano) and shows how far a model ranking depends on the evaluation window and output version (archived UK services-inflation example, 3- and 6-month horizons, pseudo-out-of-sample design with look-ahead documented).

Research Fields

Empirical asset pricing (cross-sectional return prediction, characteristic-based factor models); machine learning and big data in finance (regularised and latent-factor estimators on large characteristic panels); financial econometrics (identification and inference in high-dimensional factor models); portfolio choice and risk management.

Education

Aug 2019–Aug 2026 — **Imperial College London**, Imperial Business School, Department of Finance, London
PhD in Finance, conferred August 2026
Supervisors: Prof. Paolo Zaffaroni (main) and Prof. Pasquale Della Corte
Thesis: No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing. Examined March 2026.
MRes in Finance, with Distinction, 2020

Aug 2017–Feb 2019 — **Columbia University**, Graduate School of Arts and Sciences, New York
M.A. in Economics (Financial Econometrics; STEM-designated programme)
Adviser: Prof. Jushan Bai. Master's thesis: Diffusion Index Forecasting: A Robust Estimator for Common Shock

Aug 2014–Apr 2017 — **University of Michigan**, Ann Arbor
B.S., double major in Economics and Mathematics (Mathematics of Finance and Risk Management track)

Research

Working Papers

Characteristic Libraries and Portfolio Decisions

Working paper, September 2026

Copying one theme or rescaling themes (theme balancing) adds no information, yet on 153 U.S. characteristics both shift attribution and holdings in four tuned ridge-type procedures; balancing makes same-date target weights differ by 19–28% of average gross exposure. Payoffs depend on method, costs and missing-return valuation. Monthly ridge's 39-to-153 expansion has small, imprecise predictive effects.

PDF

Characteristic-Space Metrics in Factor Models: Identification and Inference

SSRN preprint, September 2026. DOI: 10.2139/ssrn.7445120

Factor models with identical fit can credit returns differently to latent factors, observed factors and an intercept; fit cannot choose the split, but the inner product chosen for characteristic space does. The paper derives rotation-invariant inference that carries the Gram metric's estimation error, and drift-robust confidence intervals near an intercept cap (exposure budget).

PDF · Online supplement · SSRN

A Geometric Framework for Identification in Characteristic-Based Factor Models

SSRN preprint, March 2026 (preliminary version). DOI: 10.2139/ssrn.7013178. Grows out of the PhD thesis; its inference theory is developed further in Characteristic-Space Metrics.

Using a metric taken from the long-run cross-sectional distribution of firm characteristics, the paper derives NA-IPCA, which nests IPCA, caps how much the intercept may carry and admits rotation-invariant inference. In the headline calibrated simulation, NA-IPCA recovers the latent pricing span and an intercept-free IPCA benchmark does not; that design does not isolate the metric.

PDF · SSRN

Characteristic Geometry and Portfolio Choice

Working paper, September 2026

On 132 U.S. characteristics, Euclidean and Gram-metric IPCA make nearly identical forecasts but different portfolios: a fixed factor-covariance ridge penalises exposures differently; removing only that ridge shrinks the largest relative weight gap to about 0.01%. In one NA-IPCA fit, using the intercept raises the gross Sharpe ratio; the mean-return gain since October 2007 is imprecise.

PDF · Slides (PDF)

Interpreting Estimated Pricing Errors: Evidence from Characteristic-Based Return Forecasts

Working paper, September 2026

What does a smaller estimated pricing error (alpha) tell a model user? On 153 U.S. characteristics, crossing the expected-return estimates and factor spaces of one model's equal- and value-weighted fits shows that their gap in fitted intercept magnitudes mainly reflects different estimated means; residual diagnostics add little to forecast selection (exploratory evidence).

PDF

PhD Thesis

No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing

PhD thesis, Imperial College London; degree conferred August 2026. Thesis available on request.

Extends IPCA, via the empirical characteristic metric, to nested QZ-IPCA and NA-IPCA models separating latent, observable (Fama–French five factors plus momentum) and residual-pricing components under an intercept cap; gives identification results, an asymptotic linear expansion and projected alternating least squares; simulations and a Jensen–Kelly–Pedersen U.S. panel show near-identical fit coexisting with different identified pricing directions.

Positions

Jun 2026–present

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Department of Finance, Imperial Business School, Imperial College London

Host: Prof. Robert Kosowski, Co-Director of the Centre

Research agenda for the visit: economically constrained machine learning for cross-sectional asset pricing and systematic investment.

- Machine learning and high-dimensional data: extending the No-Arbitrage IPCA framework of the thesis to regime-switching factor models in which risk premia vary with macro conditions; combining penalised estimation with economically constrained representation learning to separate alpha signals that survive those constraints from spurious signals driven by weak factors or data snooping.
- Research and practice: using the quantitative platform of the Imperial Student Investment Fund to test research signals as implementable strategies, net of transaction costs and within turnover and risk limits.

Selected engagements

Independent Quantitative Research & AI/ML Consultant, project-based advisory engagements

- Advised emerging systematic investment teams in China on quantitative research infrastructure, data and feature engineering, and Python-based research workflows.

- Helped teams adopt cloud-based AI coding agents and multi-agent workflows for research-platform development, model implementation, testing and research automation, with the necessary human review kept in place.

In a personal capacity, separate from the Imperial appointment. Scope and terms: mingyangliu.org/advisory

Research Experience

Aug 2019–Aug 2026 **Doctoral research**, Department of Finance, Imperial College London (asset pricing, financial econometrics, machine learning in finance)

- Models and geometry: characteristic-based factor models with an explicit characteristic-space geometry, using the empirical Gram metric of firm characteristics as the identification convention; the extension of IPCA to the nested QZ-IPCA and NA-IPCA models developed in the thesis.
- Identification and inference: rotation-invariant identification (loading classes modulo latent rotations); asymptotic linear representations that carry the metric's estimation error into standard errors; drift-robust confidence intervals near an intercept cap (exposure budget); projected alternating least-squares estimation; geometry-calibrated Monte Carlo designs.
- Portfolio experiments and trading costs: rolling out-of-sample portfolios (120-month windows, U.S. stocks 1973–2024) that trace weight gaps across metrics to a fixed factor-covariance ridge; information-preserving library interventions (exact copies, theme balancing) across four tuned ridge-type procedures, traced to theme attribution, stock-level ledgers and returns under transaction-cost scenarios; a separate 39-to-153 characteristic expansion of monthly ridge evaluated for stock-return prediction; forecast-evaluation designs that cross expected-return estimates with factor spaces.
- Data and tools: the Jensen–Kelly–Pedersen library of 153 U.S. stock characteristics (jkpfactors.com; WRDS `contrib.global_factor`, built from CRSP/Compustat), 1962–2025, with point-in-time formation alignment; a CRSP/Compustat panel built following the Kelly–Pruitt–Su replication dataset (36 characteristics plus a constant, 12,813 firms, 1964–2014); CRSP daily dollar volume for trading-capacity checks; NBER recession dates. Author of `naipca`, a Python toolkit implementing IPCA, QZ-IPCA and NA-IPCA with a companion metric-diagnostics package (available to academic users on request).

2026 **Research Toolkit demo** (independent research; archived example): built the forecast-evaluation workflow shown in Macro Forecast Evaluation & Regime Diagnostics.

May 2018–May 2019 **Research Assistant (macro forecasting)**, Department of Economics, Columbia University

- Diffusion-index and factor models that extract common inflation and interest-rate factors from large mixed-frequency macro data; robust PCA and stable principal component pursuit for missing data, revisions, outliers and structural breaks.

Jun 2016–Jan 2017 **Research Assistant (derivatives pricing)**, Department of Mathematics, University of Michigan

- Real-option valuation of a gold mine, using the Lihir Gold Limited IPO as a case: estimated gold-price volatility and the present value of total costs, took strike parameters from the prospectus, and priced the option with a binomial tree.

Teaching

2023, 2024 **Graduate Teaching Assistant**, Imperial Business School

Financial Statistics, MSc Risk Management & Financial Engineering. Student rating 4.8 out of 5.

Nominated by students for the Excellence in Teaching Award (both years).

Teaching notes (PDF, 96 pp., 15 MB, shared for academic use):
Teaching notes: Financial Statistics

Scholarships and Honours

2019–2025 Fully funded doctoral scholarship, Imperial Business School (Graduate Teaching Assistant Scholarship)

2014–2016 University Honors, University of Michigan (three terms)

Service

May 2024–present

Adviser; former Chair of the Executive Committee and Advisory Committee, Imperial Student Investment Fund

- Helped the Head of the Department of Finance set up a lab for the Student Investment Fund, linking teaching, research and industry practice.
- Helped build the fund's quantitative research platform, including a Python multi-asset research and trading framework covering data, signals, backtesting, transaction costs, portfolio accounting, risk control and reporting.
- Helped bring in the Trading and Risk Management programme of Qube Research & Technologies (QRT) and portfolio-optimisation training on LSEG Workspace.
- Mentored the fund's student teams in their first QRT Quantitative Competition (2026; four teams reached the final, two placed second and third).

Skills and Languages

Programming	Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, PyTorch), MATLAB, R, SQL, Git
Methods	Cross-sectional asset pricing (IPCA, QZ-IPCA, NA-IPCA, Fama–MacBeth regressions), latent-factor models, regularised regression and portfolio shrinkage (ridge-type rules, UPSA), robust PCA (RPCA, SPCP), moving-block bootstrap inference, Monte Carlo simulation design, neural networks and transfer learning, state-space and regime-switching models, financial text analysis
Data	CRSP/Compustat and the Jensen–Kelly–Pedersen characteristic library (jkpfactors.com) via WRDS; Bloomberg; LSEG Workspace
Languages	Chinese (Mandarin, native); English (fluent; working language)

References

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Updated October 2026